added. As you would hope, the third complex number represents the third sinusoid. The complex addition matches the physical system.

Now, imagine multiplying the complex numbers *A* and *B*, resulting in another complex number. Does this match what happens when the two sinusoids are multiplied? No! Multiplying two sinusoids does *not* produce another sinusoid. Complex multiplication fails to match the physical system, and therefore cannot be used.

Fortunately, the valid operations are clearly defined. Two conditions must be satisfied. First, all of the sinusoids must be at the *same frequency*. For example, if the complex numbers: $1+1j$ and $2+2j$ represent sinusoids at the same frequency, then the sum of the two sinusoids is represented by the complex number: $3+3j$. However, if $1+1j$ and $2+2j$ represent sinusoids with different frequencies, there is nothing that can be done with the complex representation. In this case, the sum of the complex numbers, $3+3j$, is meaningless.

In spite of this, frequency can be left as a variable when using complex numbers, but it must be the *same* frequency everywhere. For instance, it is perfectly valid to add: $2\omega+3\omega j$ and $3\omega+1j$, to produce: $5\omega+(3\omega+1)j$. These represent sinusoids where the amplitude and phase vary as frequency changes. While we do not know *what* the particular frequency is, we do know that it is the *same* everywhere, i.e., ω.

The second requirement is that the operations being represented must be *linear*, as discussed in Chapter 5. For instance, sinusoids can be combined by addition and subtraction, but not by multiplication or division. Likewise, systems may be amplifiers, attenuators, high or low-pass filters, etc., but not such actions as: squaring, clipping and thresholding. Remember, even convolution and Fourier analysis are only valid for linear systems.

Complex Representation of Systems

Figure 30-4 shows an example of using complex numbers to represent a sinusoid passing through a linear system. We will use continuous signals for this example, although discrete signals are handled the same way. Since the input signal is a sinusoid, and the system is linear, the output will also be a sinusoid, and at the same frequency as the input. As shown, the example input signal has a conventional representation of: $3\cos(\omega t+\pi/4)$, or the equivalent expression: $2.1213\cos(\omega t)-2.1213\sin(\omega t)$. When represented by a complex number this becomes: $3e^{-j\pi/4}$ or $2.1213+j2.1213$. Likewise, the conventional representation of the output is: $1.5\cos(\omega t-\pi/8)$, or in the alternate form: $1.3858\cos(\omega t)+0.5740\sin(\omega t)$. This is represented by the complex number: $1.5e^{j\pi/8}$ or $1.3858-j0.5740$.

The system's characteristics can also be represented by a complex number. The magnitude of the complex number is the ratio between the magnitudes